

- 13** A mutation occurred within the DNA sequence coding for an enzyme, causing a decrease in the rate of a reaction catalysed by this enzyme.

Which statements could explain the decrease in the rate of reaction?

- 1 An inhibitor for this enzyme has an increased affinity for the enzyme and forms an enzyme–inhibitor complex more easily.
- 2 The active site of the enzyme might have changed shape and so is no longer complementary.
- 3 The activation energy for the reaction with the mutated enzyme is greater than with the non-mutated enzyme.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only